

World Map

Mr. Pacha, umu archeologist wo muri Bolivian, yavumbuye document ya kera hafi ya Tiwanaku isobanura isi muri period ya tiwanaku (300-1000 CE). Icyo gihe, hari hari ibihugu N , bitondekanyijwe kuva kuri 1 kugera kuri N .

Muri icyo document, hari list ya M different pairs z' ibihugu byegeranye:

$$(A[0], B[0]), (A[1], B[1]), \dots, (A[M-1], B[M-1]).$$

Kuri buri i ($0 \leq i < M$), document ivuga ko igihugu $A[i]$ cyari cyegeranye n' igihugu $B[i]$ and vice versa. Amatsinda y' ibihugu atavuzwe ntago yari yegeeranye

Mr. Pacha arashaka gukora map y' isi ku buryo uburyo ibihugu byegeranye biba bimeze kimwe nkuko byari bimeze mu bihe bya Tiwanaku. Kubera ibyo, yatoranyije positive integer K . Hanyuma, ashushanya map nka grid of $K \times K$ square cells, aho rows zitondekanyijwe kuva kuri 0 kugera kuri $K-1$ (hejuru kugera hasi) kandi columns zitondekanyijwe kuva kuri 0 kugera kuri $K-1$ (ibumoso kugera iburyo).

Arashaka gushyira ibara muri buri cell ya map akoresheje rimwe mu mabara N . Amabara atondekanyijwe kuva kuri 1 kugera kuri N , kandi igihugu j ($1 \leq j \leq N$) gihagarariwe n' ibara j . Amabara **agomba gukurikiza ibi bikurikira**:

- Kuri buri j ($1 \leq j \leq N$), hari nibura one cell ifite ibara j .
- Kuri buri pair y' ibihugu byegeranye $(A[i], B[i])$, hari nibura one pair of adjacent cells aho imwe muri zo ifite ibara $A[i]$ kandi indi ifite ibara $B[i]$. Cells ebyiri ziba zegeeranye mu gihe zihuriye ku ruhade.
- Kuri buri pair ya adjacent cells ifite different colors, ibihugu biri represented n' izo two colors byari byegeranye mu bihe bya Tiwanaku.

Urugero, igihe $N = 3$, $M = 2$ kandi pairs z' ibihugu byegeranye ari $(1, 2)$ na $(2, 3)$, bivuze ko pair $(1, 3)$ zitari zegeeranye, kandi iyi map ifite dimension $K = 3$ yujuje conditions zose.

2	3	3
2	3	2
1	2	1

Muri make, igihugu **ntago** kigomba gukora connected region kuri map. Kuri map iri haruguru, igihugu cya 3 gikora connected region, mu gihe ibihugu 1 na 2 bikora disconnected regions.

Akazi ufite ni ugufasha Mr. Pacha gutora umubare K hanyuma agakora map. Document yizeza ko uwo mubare ubaho. Kubera ko Mr. Pacha akunda maps ntoya, muri subtask yanyuma score yawe idependinga kuri value ya K , na lower values za K zishobora gutanga better score. Gusa, gushaka umubare muto ushoboka wa K ntago ari ngombwa.

Implementation Details

Ugomba gukora ibi bikurikira:

```
std::vector<std::vector<int>> create_map(int N, int M,
std::vector<int> A, std::vector<int> B)
```

- N : umubare w' ibihugu.
- M : umubare w' ibihugu byegeranye.
- A na B : arrays ifite length M isobanura ibihugu byegeranye.
- Ibi bikorwa inshoro zitarenze 50 kuri buri test case.

Iyi function ireturninga an array C yerekana map. Reka K ibe length ya C .

- Buri element of C igomba kuba an array of length K , irimo integers ziri hagati ya 1 na N inclusive.
- $C[i][j]$ n' ibara rya cell riri muri row i na column j (for each i and j such that $0 \leq i, j < K$).
- K igomba kuba muni cyangwa ingana nka 240.

Constraints

- $1 \leq N \leq 40$
- $0 \leq M \leq \frac{N \cdot (N-1)}{2}$
- $1 \leq A[i] < B[i] \leq N$ kuri buri i aho $0 \leq i < M$.

- Pairs $(A[0], B[0]), \dots, (A[M-1], B[M-1])$ ziratandukanye.
- Hari nibura map imwe yujuje ibisabwa.

Subtasks and Scoring

Subtask	Score	Additional Constraints
1	5	$M = N - 1, A[i] = i + 1, B[i] = i + 2$ for each $0 \leq i < M$.
2	10	$M = N - 1$
3	7	$M = \frac{N \cdot (N-1)}{2}$
4	8	Igihugu 1 kegeranye n' ibihugu byose. Amwe mu matsinda y' ibihugu ashobora kuba nayo yegeranye.
5	14	$N \leq 15$
6	56	Ntazindi constraints.

Muri subtask 6, score yawe idependinga kuri value ya K .

- Igihe map yabaye returned by `create_map` itujuje ibisabwa, score yawe kuri subtask izaba 0.
- Bitabaye ibyo, reka R ibe value **nini ishoboka ya K/N** muri calls zose za `create_map`. Then, ubona **partial score** hakurikijwe iyi table:

Limits	Score
$6 < R$	0
$4 < R \leq 6$	14
$3 < R \leq 4$	28
$2.5 < R \leq 3$	42
$2 < R \leq 2.5$	49
$R \leq 2$	56

Example

Muri CMS, zino scenarios zose ziba zikuboye muri test case imwe.

Example 1

Tekereza uhamagaye iyi function:

```
create_map(3, 2, [1, 2], [2, 3])
```

Uru ni urugero ruturutse muri task description, bivuze ko function izareturning iyi map.

```
[
  [2, 3, 3],
  [2, 3, 2],
  [1, 2, 1]
]
```

Example 2

Tekereza guhamagara iyi function:

```
create_map(4, 4, [1, 1, 2, 3], [2, 3, 4, 4])
```

Muri uru rugero, $N = 4$, $M = 4$ n' amatsinda y' ibihugu (1,2), (1,3), (2,4), na (3,4) biregeranye. Bityo rero, pairs (1,4) na (2,3) ntago zegeranye.

Iyi function ishobora gutanga iyi map ifite dimension $K = 7$, yujuje ibisabwa byose.

```
[
  [2, 1, 3, 3, 4, 3, 4],
  [2, 1, 3, 3, 3, 3, 3],
  [2, 1, 1, 1, 3, 4, 4],
  [2, 2, 2, 1, 3, 4, 3],
  [1, 1, 1, 2, 4, 4, 4],
  [2, 2, 1, 2, 2, 4, 3],
  [2, 2, 1, 2, 2, 4, 4]
]
```

Map ishobora kuba ntoya; urugero, function ishobora gutanga map ifite dimension $K = 2$.

```
[
  [3, 1],
  [4, 2]
]
```

Mpenya ko izo map zose zubahirije condition $K/N \leq 2$.

Sample Grader

Umurogo wambere wa input ungomba kuba single integer T , umubare wa scenarios. Description ya T scenarios igomba gukurikira, buri imwe muri format yerekanywe hano:

Input Format:

```
N M
A[0] B[0]
:
A[M-1] B[M-1]
```

Output Format:

```
P
Q[0] Q[1] ... Q[P-1]

C[0][0] ... C[0][Q[0]-1]
:
C[P-1][0] ... C[P-1][Q[P-1]-1]
```

Hano, P ni length of the array C yareturninzwe na `create_map`, kandi $Q[i]$ ($0 \leq i < P$) ni length ya $C[i]$. Mpenya ko umurongo wa 3 muri output ntakintu kirimo ku bushake.